Machine Learning
Assignment 2: The Morning Rush

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Google Colab Link

- Notebook (Parts II and III): `https://colab.research.google.com/drive/14agXEi8Io2W_C5021Mbf59dkZ5SfQX3I?usp=sharing`

All code runs on the free Colab CPU. Every scaler and encoder is placed inside a scikit-learn `Pipeline` (through a `ColumnTransformer`), so it is fitted on the training rows only. All random seeds are fixed to 0. Only `LinearRegression`, `Ridge`, `SGDRegressor` and `PolynomialFeatures` are used.

1 Part I: Least Squares by Hand

1.1 (a) OLS and Ridge

Deriving the ridge solution

The ridge objective is

$$J_{\text{ridge}}(w) = \tfrac{1}{2}\|y - Xw\|_2^2 + \tfrac{\lambda}{2}\|w\|_2^2 = \tfrac{1}{2}\left(y^\top y - 2w^\top X^\top y + w^\top X^\top X w\right) + \tfrac{\lambda}{2} w^\top w.$$

Its gradient is

$$\nabla_w J_{\text{ridge}} = -X^\top y + X^\top X w + \lambda w = -X^\top (y - Xw) + \lambda w.$$

Setting $\nabla_w J_{\text{ridge}} = 0$ gives the normal equations

$$(X^\top X + \lambda I)\, w = X^\top y \quad \Longrightarrow \quad w^* = (X^\top X + \lambda I)^{-1} X^\top y.$$

Uniqueness for every $\lambda > 0$. For any $v \neq 0$,

$$v^\top (X^\top X + \lambda I) v = \|Xv\|_2^2 + \lambda \|v\|_2^2 \;\geq\; \lambda \|v\|_2^2 > 0.$$

Hence $X^\top X + \lambda I$ is positive definite. All of its eigenvalues are at least $\lambda > 0$, so it is invertible and w^* exists. The Hessian $\nabla_w^2 J_{\text{ridge}} = X^\top X + \lambda I$ is positive definite, so J_{ridge} is strictly convex and w^* is its unique minimiser.

This holds even when $X^\top X$ is singular, for example when features are collinear or $N < d$. A singular $X^\top X$ has some zero eigenvalues, so OLS has infinitely many minimisers. Adding λI shifts every eigenvalue up by λ, which removes the zero eigenvalues.

OLS on the toy dataset

With $x = (0, 1, 2, 3, 4)$ and $y = (10, 8, 7, 4, 1)$ we have $\bar{x} = 2$ and $\bar{y} = 30/5 = 6$.

$x^{(i)}$	$y^{(i)}$	$x^{(i)}-\bar{x}$	$y^{(i)}-\bar{y}$	$(x^{(i)}-\bar{x})^2$	$(x^{(i)}-\bar{x})(y^{(i)}-\bar{y})$
0	10	-2	4	4	-8
1	8	-1	2	1	-2
2	7	0	1	0	0
3	4	1	-2	1	-2
4	1	2	-5	4	-10
				$S_{xx}=10$	$S_{xy}=-22$

$$w_1=\frac{S_{xy}}{S_{xx}}=\frac{-22}{10}=-2.2, \qquad w_0=\bar{y}-w_1\bar{x}=6-(-2.2)(2)=10.4.$$

The fitted line is $\hat{y}=10.4-2.2x$.

x	0	1	2	3	4
$\hat{y}$	10.4	8.2	6.0	3.8	1.6
$y-\hat{y}$	-0.4	-0.2	1.0	0.2	-0.6
$(y-\hat{y})^2$	0.16	0.04	1.00	0.04	0.36

$$\text{SSE}=1.6, \qquad \text{MSE}=\frac{1.6}{5}=0.32, \qquad \text{SST}=\sum_i (y^{(i)}-\bar{y})^2=16+4+1+4+25=50,$$

$$R^2=1-\frac{\text{SSE}}{\text{SST}}=1-\frac{1.6}{50}=0.968.$$

Ridge on the toy dataset

The intercept is not penalised, so $w_1=S_{xy}/(S_{xx}+\lambda)$ and $w_0=\bar{y}-w_1\bar{x}$.

- $\lambda=1$: $w_1=\dfrac{-22}{10+1}=-2.0$ and $w_0=6+2.0\cdot 2=10.0$. The predictions are $(10,8,6,4,2)$, so MSE $=(0+0+1+0+1)/5=0.40$.
- $\lambda=10$: $w_1=\dfrac{-22}{10+10}=-1.1$ and $w_0=6+1.1\cdot 2=8.2$. The predictions are $(8.2,7.1,6.0,4.9,3.8)$, so MSE $=13.7/5=2.74$.

As $\lambda\to\infty$. $w_1=-22/(10+\lambda)\to 0$ monotonically: $-2.2\to-2.0\to-1.1\to\cdots\to 0$. The intercept tends to $\bar{y}=6$, so the model collapses to the constant prediction $\hat{y}=6$. The training MSE grows towards the variance of y, which is $50/5=10$. Larger λ trades fit (bias) for a smaller, more stable slope (variance).

1.2 (b) Units Matter

The new feature is $x'=x/10$ (cm): $x'=(0,0.1,0.2,0.3,0.4)$ and $\bar{x}'=0.2$. In general, for $x'=x/c$,

$$S'_{xx}=\frac{S_{xx}}{c^2}=\frac{10}{100}=0.1, \qquad S'_{xy}=\frac{S_{xy}}{c}=\frac{-22}{10}=-2.2.$$

OLS.

$$w_1' = \frac{S_{xy}'}{S_{xx}'} = \frac{-2.2}{0.1} = -22 \text{ per cm}, \qquad w_0' = 6 - (-22)(0.2) = 10.4.$$

The predictions are

$$\hat{y} = w_0' + w_1' x' = 10.4 - 22 \cdot \frac{x}{10} = 10.4 - 2.2x,$$

which are identical to part (a). In general $w_1' = c\, w_1$, so $w_1' x' = w_1 x$. OLS compensates exactly for any rescaling, and the predictions, MSE and R^2 are unchanged.

Ridge with $\lambda = 1$.

$$w_1' = \frac{S_{xy}'}{S_{xx}' + \lambda} = \frac{-2.2}{0.1 + 1} = -2.0 \text{ per cm } = -0.2 \text{ per mm}.$$

In millimetres (part a) the ridge slope was -2.0 per mm. After switching to centimetres it is only -0.2 per mm, which is ten times smaller.

Why ridge is not scale invariant. For $x' = x/c$,

$$\frac{w_1'}{c} = \frac{S_{xy}/c^2}{S_{xx}/c^2 + \lambda} = \frac{S_{xy}}{S_{xx} + \lambda c^2}.$$

Measuring precipitation in cm is equivalent to using $\lambda c^2 = 100$ in the original units. To produce the same prediction, the cm-coefficient must be $10\times$ larger, and the penalty $\frac{\lambda}{2} w^2$ charges it $100\times$ more. The penalty depends on the units of each feature, so the amount of shrinkage depends on arbitrary measurement choices. This is why features should be standardised before using ridge or lasso.

Why gradient descent needs scaling. The Hessian of the least-squares loss with an intercept column is $X^\top X$.

- mm: $X^\top X = \begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix}$, with eigenvalues 33.51 and 1.49, so the condition number is $\kappa \approx 22.5$.
- cm: $X^\top X = \begin{bmatrix} 5 & 1 \\ 1 & 0.3 \end{bmatrix}$, with eigenvalues 5.20 and 0.096, so $\kappa \approx 54.2$.
- standardised (centred, unit variance): $X^\top X \propto I$, so $\kappa = 1$.

Gradient descent with step η is stable only if $\eta < 2/\lambda_{\max}$. The error along the flattest direction shrinks by a factor of $|1 - \eta\lambda_{\min}|$ per step, so the number of iterations grows roughly in proportion to $\kappa = \lambda_{\max}/\lambda_{\min}$.

- With features on very different scales, a step size small enough for the steep direction makes almost no progress along the flat direction, so convergence is slow.
- A step size tuned for typical (unit-scale) features exceeds $2/\lambda_{\max}$ along a large-scale feature, so the iterates diverge.

Standardising makes the loss surface close to spherical, so one step size works for all directions. Part II(c) shows this effect on the real data.

2 Part II: Feature Engineering and Scaling

Data and split. `train.csv` has 108,720 station-hours (30 stations × 3,624 hours, Jan–May 2026). The validation set is the last month (May, 22,320 rows), and the training set is Jan–Apr (86,400 rows). RMSE and R^2 are always computed on the original rental scale, after clipping negative predictions to 0.

2.1 (a) Exploration and Baseline

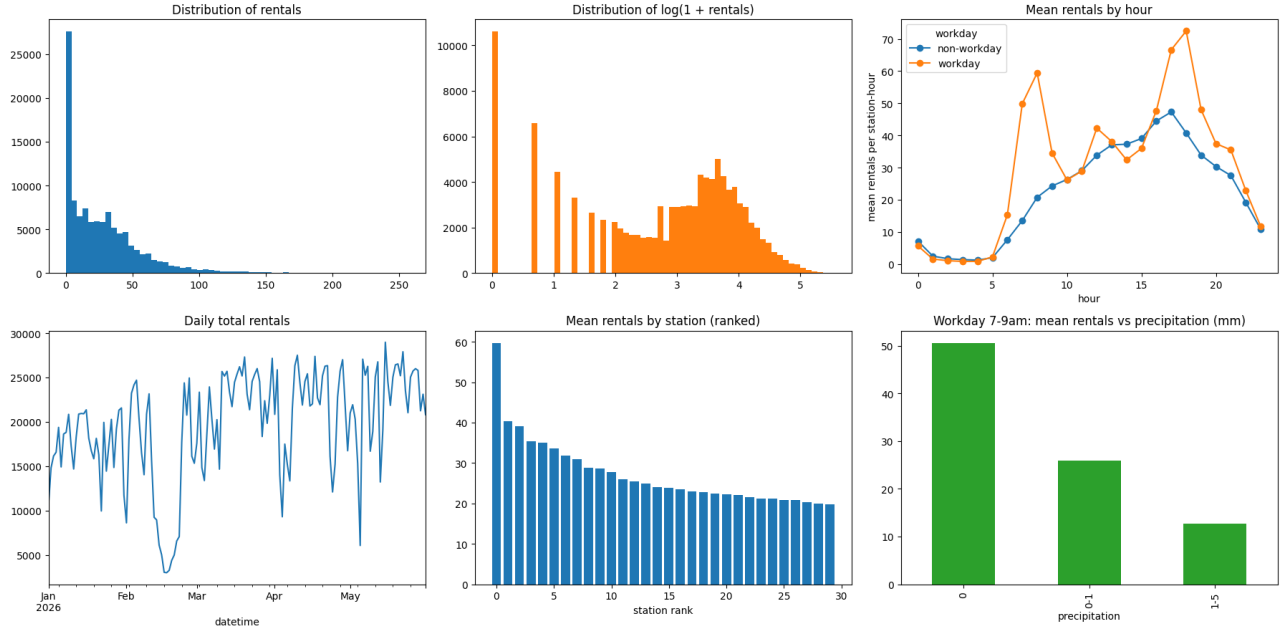


Figure 1: Target distribution, hourly profile, daily totals, station means, and the effect of rain on workday morning-peak demand.

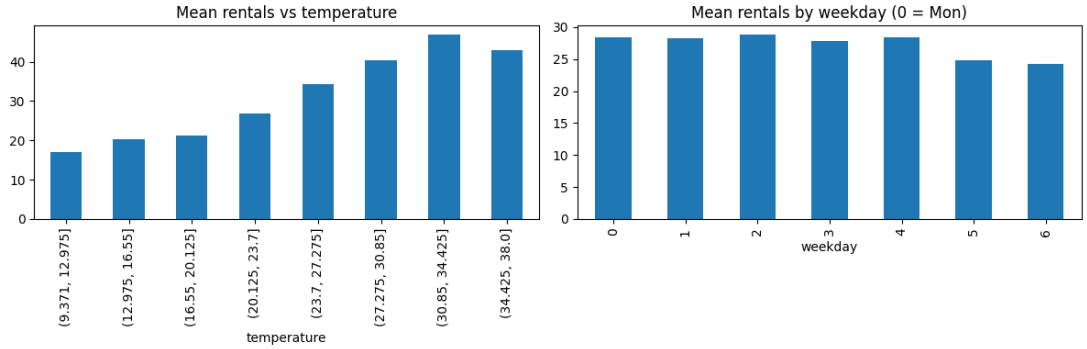


Figure 2: Mean rentals by temperature bin and by day of week.

Main patterns.

1. **Time of day × workday is the dominant signal.** On workdays the profile has two commuting peaks: a mean of 59.4 rentals at 08:00 and 72.5 at 18:00, against 1–2 rentals between 01:00 and 04:00. On weekends and holidays there is no morning peak (20.8 at 08:00), and demand rises smoothly to a single afternoon maximum (47.3 at 17:00). The shape of the hourly curve therefore depends on whether the day is a workday.

2. **Weekdays differ from weekends.** Mon–Fri average about 28 rentals, compared with 24.8 on Saturday and 24.3 on Sunday.

3. **Stations differ strongly in level and timing.** Mean demand ranges from 59.7 (MRT Gongguan, Exit 2) to 19.9 (MRT Beitou, Exit 1). The 08:00 workday peak relative to each station's own daily mean also varies. At MRT Xiangshan (Exit 2) it is 3.96× the mean, whereas at the NTU First Student Activity Center it is only 0.22×. Commuters leave from homes and MRT exits in the morning and from campuses and offices in the evening, so each station has its own daily shape.

4. **Rain suppresses demand.** During workday 07:00–09:00, mean rentals fall from 50.6 when it is dry to 22.0 when it rains. More rain gives fewer rentals (see the precipitation bins). The raw linear correlations with the weather columns are weak ($|r| \leq 0.30$; temperature $+0.28$, humidity -0.30, precipitation -0.12), because weather effects are multiplicative and depend on the hour.

5. **Trend and holidays.** Monthly means rise from 24.2 (Jan) and 19.4 (Feb, which includes the Lunar New Year dip visible in the daily series) to 30.9–31.4 (Mar–May), together with warmer weather.

6. **The target is a skewed count.** Rentals range from 0 to 257, with mean 27.2, median 21 and skewness 1.69. 9.8% of the hours have zero rentals, and the spread grows with the level (the std per month tracks the mean).

Model	Train RMSE	Train R^2	Val RMSE	Val R^2
Baseline LinearRegression, 4 weather columns, y	**25.089**	**0.144**	**29.375**	**0.099**
Baseline LinearRegression, $\log(1+y)$	28.462	-0.102	38.159	-0.521
Ridge on feature set S3, y	14.561	0.712	16.596	0.712
Ridge on feature set S3, $\log(1+y)$	14.642	0.709	**16.262**	**0.724**

Table 1: Baseline (first row) and the effect of modelling $\log(1+y)$. RMSE is always on the original scale.

Baseline and target transformation. The baseline explains very little: validation R^2 is about 0.10, and the training RMSE matches the TA value in `baseline.txt`. Weather alone cannot represent the commuting cycle.

Decision: model $\log(1+y)$. The effects found above are multiplicative. Rain roughly halves demand at every station, and a busy MRT station has proportionally larger peaks than a quiet one. In log space these effects become additive, which is exactly the form a linear model can represent. The log transform also reduces the right skew and the heteroscedasticity, and $\exp(\cdot) - 1 \geq 0$ keeps predictions non-negative. With the expressive feature set S3, the log target lowers the validation RMSE from 16.60 to 16.26 and makes the training/validation behaviour more consistent.

The transform is not universally better. With only the four weather columns, the log-target model is worse (38.16 vs 29.37). Without time features the model predicts roughly the geometric mean, and transforming back systematically underestimates the large peak counts that dominate RMSE. The log target is useful once the features can explain where those peaks occur, which is the case for every set we use from S1 onward.

2.2 (b) Feature Engineering

The model is fixed to Ridge ($\alpha = 1$) on $\log(1+y)$, with StandardScaler on the numeric columns and OneHotEncoder (`handle_unknown="ignore"`) on the categorical columns. Every set contains all features of the previous one.

Feature set (cumulative)	#Features	Train RMSE	Val RMSE	Val R^2
S0: 4 weather columns	4	28.462	38.158	−0.521
S1: + one-hot hour (24), one-hot weekday (7), workday flag, capacity	37	20.033	22.293	0.481
S2: + one-hot station (30), hour×workday one-hot (48), rain flag	115	16.670	18.623	0.638
S3: + station×period×workday one-hot, rain×workday-peak	296	14.642	16.262	0.724
S4: + station×hour×workday one-hot	1736	**12.321**	**13.432**	**0.812**

Table 2: Cumulative feature sets with Ridge ($\alpha = 1$). Here *workday* = 1 for Mon–Fri that are not public holidays, and *period* ∈ {07–09 am peak, 17–19 pm peak, off-peak}.

What each set adds, and which helped most.

- **S1 (time features), motivated by pattern 1–2.** This was the largest single gain: validation RMSE fell from 38.2 to 22.3 and R^2 rose from −0.52 to 0.48. Hour of day is the most important variable, because it describes the commuting cycle that weather cannot.
- **S2 (station identity, hour×workday), motivated by pattern 1 and 3.** RMSE fell to 18.6. Station one-hots give each station its own baseline level, which varies three-fold between stations. The 48 hour×workday indicators let the workday curve have its 08:00/18:00 peaks while the holiday curve stays smooth. A purely additive hour effect would force both day types to share the same shape.
- **S3 (station×period, rain×peak), motivated by pattern 3 and 4.** RMSE fell to 16.3. Each station receives its own peak amplitudes, which captures morning-origin versus evening-origin stations. The rain×workday-peak term lets rain have a stronger effect on commuting hours.
- **S4 (station×hour×workday), motivated by pattern 3.** RMSE fell to 13.4 ($R^2 = 0.81$). This gives every station a full 24-hour profile for each day type. Even with 1,736 columns, the training/validation gap stays small (12.3 vs 13.4). Each indicator is supported by 43 (non-workday) to 77 (workday) training rows, and the ridge penalty shrinks each cell towards the shared hour and station effects.

Overall, the biggest improvements came from time of day (S1) and from interactions between time, day type and station (S2–S4). This shows that YouBike demand is governed mainly by *where* and *when* commuters travel, and only secondarily by weather.

Encoding choices from a linear-model perspective.

- **Hour as one-hot, not numeric.** A numeric hour forces a single slope: demand would change monotonically from 00:00 to 23:00, and 23:00 would be treated as far from 00:00. The real profile is bimodal and cyclic. One-hot encoding gives each hour a free coefficient, which can represent any shape. A sine/cosine pair would make hour cyclic but would still be too smooth to represent two sharp peaks. Part III shows how much a numeric hour underfits (Model A).
- **Weekday as one-hot.** The weekday effect is not ordinal (Saturday ≠ Friday +1), so separate offsets are appropriate.
- **Workday flag (weekend OR `is_holiday`).** Holidays falling on Mon–Fri behave like weekends (10 such days in Jan–May), so the flag captures commuting behaviour better than the calendar weekday alone.
- **Interactions as crossed one-hot categories.** A linear model is additive, so it cannot by itself make the hour effect depend on the day type or the station. Crossing the categories

(hour×workday, station×period, and station×hour×workday) builds the interaction directly into the design matrix, and Ridge controls the resulting number of parameters.

- **Station as one-hot**, because the IDs are nominal. **Capacity** is kept as a standardised numeric column; it is collinear with the station dummies, which ridge handles without difficulty.
- **Rain flag in addition to precipitation (mm).** In log space the main effect is the presence of rain (a roughly constant percentage drop), while the millimetres add a smaller graded effect.
- **Weather columns standardised** inside the pipeline. This makes the ridge penalty treat them equally (Part I(b)).

2.3 (c) Feature Scaling

The experiment uses feature set S2 (115 features) with the log target. Four preprocessing options are applied to the six numeric columns (four weather columns, capacity, rain flag); the one-hot columns are unchanged. The three models are:

- `LinearRegression`, closed form (least squares on the dense design matrix);
- `Ridge`($\alpha = 1000$), with a strong penalty so that its interaction with scale is visible;
- `SGDRegressor`(`eta0`=0.01, `max_iter`=1000, `tol`=10^{-4}, `random_state`=0), fitted by gradient descent.

Model	None	Standard	MinMax	Robust
LinearRegression (closed form)	18.624	18.624	18.624	18.624
Ridge ($\alpha = 1000$)	20.459	20.361	20.681	20.392
SGDRegressor: val RMSE	∞	18.684	18.575	18.611
SGDRegressor: GD iterations	231 (diverged)	18	17	17

Table 3: Validation RMSE for every model and scaling combination, with the number of gradient-descent epochs (`n_iter_`) for SGD.

Explanation using Part I(b).

- **LinearRegression is unaffected.** All four scalings give exactly 18.624. Scaling multiplies each numeric column by a constant and shifts it, which only reparametrises the model: $w' = c\,w$ and the intercept absorbs the shift. Exactly as for the toy OLS slope (-2.2 per mm $= -22$ per cm), the least-squares predictions are invariant.
- **Ridge changes with the scaling.** RMSE ranges from 20.36 (standard) to 20.68 (min–max). For a column scaled by $1/c$, the penalty acts like λc^2 in the original units (Part I(b)). Min–max scaling shrinks humidity (range about 30–98) and temperature to $[0, 1]$, so their coefficients must become larger, they are penalised more strongly, and the model underfits more. Without scaling, columns with a large spread (humidity, capacity) are penalised only lightly, while the precipitation column (std 0.78) is penalised relatively heavily. Standardisation gives every column the same unit variance and therefore the same penalty, which produced the lowest RMSE. The differences are modest because only six of the 115 columns are rescaled; the one-hot columns are already on a $\{0, 1\}$ scale.
- **SGD diverged without scaling.** Unscaled humidity (≈ 75), capacity (≈ 57) and temperature (≈ 21) are 1–2 orders of magnitude larger than the one-hot columns, so $X^\top X$ is extremely ill-conditioned and $\lambda_{\max}$ is huge. The step size `eta0`=0.01 exceeds $2/\lambda_{\max}$ along those directions, the weights grow without bound, and the predictions overflow to ∞. SGD stopped after 231 epochs with non-finite coefficients and never converged. After any of the three scalings, the

condition number is small enough that the same step size converges in 17–18 epochs to a solution within 0.06 RMSE of the closed-form one. This is the κ argument of Part I(b): scaling does not change the optimum, but it decides whether gradient descent can reach it.

3 Part III: Learning Curves and Forecasting

Learning curves use all of `train.csv` (Jan–May, 108,720 rows) with 5-fold unshuffled `KFold`, so each validation fold is a contiguous block of about one month. There are eight training sizes from 5% to 100% of each fold's training part, and the scoring metric is RMSE on the original scale.

3.1 (a) Diagnosis

- **Model A (deliberately simple):** `LinearRegression` on the four weather columns plus `hour` as *one numeric column*, with standardised inputs and the log target (5 features). The model can only express a monotone hourly trend.
- **Model B (deliberately flexible):** one-hot station, hour×workday and weekday, plus the four weather columns, all expanded with `PolynomialFeatures(degree=2)` to every pairwise product (4,094 features). It is fitted with *unregularised* `LinearRegression`.

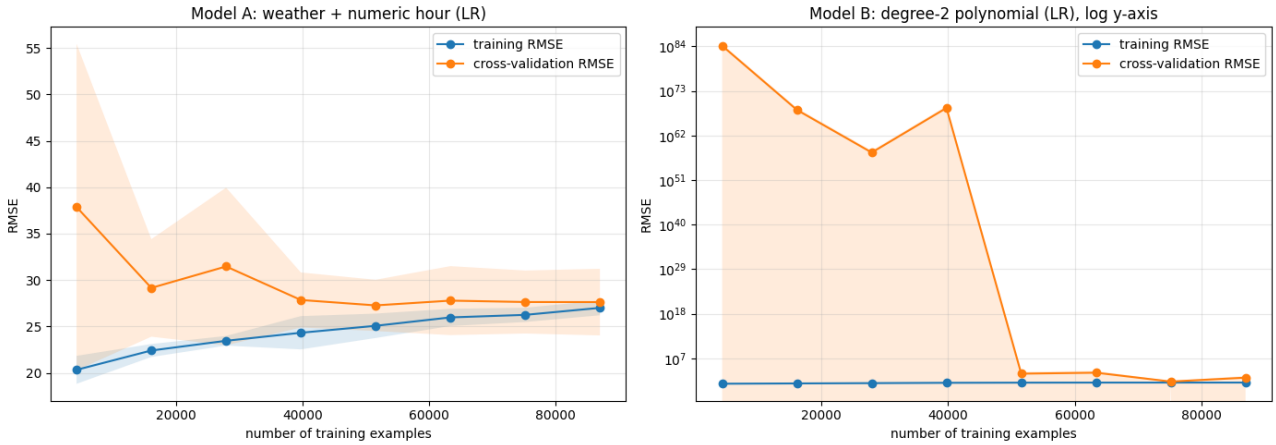


Figure 3: Learning curves of Model A (left) and Model B (right; log y-axis because the CV error spans many orders of magnitude). Shaded bands show ±1 std over folds.

Model	Final train RMSE	Final CV RMSE	Gap	May val RMSE	Diagnosis
A	27.01	27.63	0.62	31.89	High bias
B	10.64	172.02	161.39	337.62	High variance

Model A: high bias (underfitting).

- At full size the training RMSE (27.0) and CV RMSE (27.6) are both high, close to the weather-only baseline and twice the error of the S4 model (13.4).
- The gap between them is small (0.6).
- As n grows, the training error rises and the CV error falls, and both curves flatten out from about 40,000 examples onwards, having converged to the same high level.

Collecting more data would not help. The curves have already met, and the remaining error comes from the model's inability to represent the bimodal, station- specific commuting cycle.

Model B: high variance (overfitting).

- The training RMSE is the lowest of all models (10.6) and almost flat across sizes.
- The CV RMSE is enormous: about 10^{84} with 4,300 examples (fewer rows than features) and still 172 at full size, giving a final gap of 161.
- Unregularised least squares on 4,094 nearly collinear products of indicator variables fits the training noise and extrapolates wildly in rarely seen combinations. The log target amplifies these errors, because a large error in $\log(1+y)$ becomes a huge error after $\exp(\cdot)-1$.

More data helps, but a large gap remains. The CV RMSE decreases substantially as n increases. However, at the largest training size, the CV RMSE is still about 172, compared with a training RMSE of 10.6, leaving a gap of about 161. This indicates persistent high variance. Collecting more data may help further, but the learning curves do not establish how much additional data would be needed. Regularization is therefore a practical remedy.

3.2 (b) Remedies

- **Model A (high bias) → richer features.** Hour is replaced by the S3 feature set (one-hot hour/weekday/station, hour×workday, station×period×workday, rain×peak; 296 features), fitted with Ridge ($\alpha = 1$).
- **Model B (high variance) → regularisation with a tuned strength.** The same 4,094 polynomial features are kept, but `LinearRegression` is replaced by Ridge. The value of α is chosen on the May validation set.

α	1	10	100	1000	**3000**	10000	30000
Train RMSE	10.86	11.13	13.31	17.35	20.46	24.05	26.35
Val RMSE	297.76	155.03	64.32	21.28	**20.52**	24.51	27.83

Table 4: Ridge strength search for Model B. Train is Jan–Apr and validation is May.

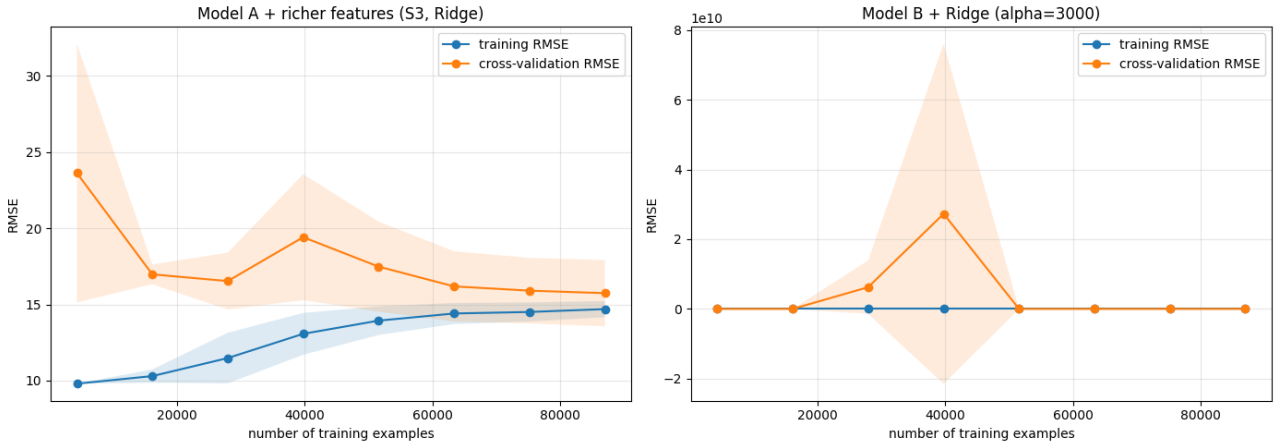


Figure 4: Learning curves after the remedies. Right: the spike at 28k–40k examples ($\sim 3 \times 10^{10}$) comes from the April and May folds only; from 51k examples onward the CV RMSE stays near 20–23 (see text).

Model	Final train RMSE	Final CV RMSE	Gap	May val RMSE
A + richer features (S3, Ridge)	14.69	15.74	1.05	16.26
B + Ridge ($\alpha = 3000$)	20.90	21.35	0.46	20.52

Did the remedies work?

- **Model A: yes.** Both curves dropped by about 12 RMSE (train 27.0 → 14.7, CV 27.6 → 15.7), and the gap remains small (1.0). This is the expected signature of fixing bias: the whole curve moves down rather than the gap closing. The validation RMSE nearly halves (31.9 → 16.3).
- **Model B: yes at full data size, with one caveat at medium sizes.**
  - At full size the gap shrank from 161 to 0.46, the CV RMSE fell from 172 to 21.4, and the May validation RMSE fell from 338 to 20.5.
  - The training error increased (10.6 → 20.9), as expected when a regulariser is added.
  - The redrawn curve still shows a spike of about 3×10^{10} at 28k–40k examples. Inspecting the folds shows that it comes only from the two folds whose validation blocks are April and May. With unshuffled `KFold`, their first 28k–40k training rows cover only January–February, where the maximum temperature is 29 °C, while the validation blocks reach 32.9 °C and 38 °C. The squared weather terms extrapolate outside the training range, the log-space prediction becomes large, and $\exp(\cdot) - 1$ magnifies it.
  - Once the training rows include spring data ($n \geq 51$k), all five folds stay between 18 and 28 RMSE.

  Ridge therefore removed the variance coming from the 4,094 nearly collinear indicator products. It cannot prevent extrapolation of polynomial terms in continuous features outside the observed range, which is a further argument against generic polynomial expansion for this task.

  The search table shows the bias–variance trade-off directly. Small α leaves the variance problem (val RMSE 298), large α underfits (val RMSE 27.8), and the optimum is $\alpha = 3000$. The strong penalty needed to stabilise the expansion makes Model B somewhat biased. Its validation RMSE (20.5) is worse than that of models with well-designed interactions (S3: 16.3, S4: 13.4). Targeted feature engineering is therefore a better use of model capacity than a generic polynomial expansion.

3.3 (c) Leaderboard

Model selection (validation set only). Thirteen candidates were compared on the May validation set: S2/S3/S4 with Ridge $\alpha \in \{0.3, 1, 3, 10\}$, and the tuned Model B.

Candidate	#Features	Train RMSE	Val RMSE	Val R^2
S4 + Ridge($\alpha = 0.3$)	1736	12.308	**13.425**	**0.812**
S4 + Ridge($\alpha = 1$)	1736	12.321	13.432	0.812
S4 + Ridge($\alpha = 3$)	1736	12.359	13.456	0.811
S4 + Ridge($\alpha = 10$)	1736	12.503	13.574	0.808
S3 + Ridge($\alpha = 0.3$)	296	14.639	16.259	0.724
S2 + Ridge($\alpha = 0.3$)	115	16.669	18.623	0.638
Model B + Ridge($\alpha = 3000$)	4094	20.458	20.524	0.560

Table 5: Best candidates (full table in `part3c_candidates.csv`).

Final model. The final model is S4 + Ridge($\alpha = 0.3$) on $\log(1 + y)$, with StandardScaler and OneHotEncoder inside the pipeline. It was retrained on all of `train.csv` (Jan–May). Predictions for `test.csv` (June) were transformed back with $\exp(\cdot) - 1$, negative values were clipped to 0, and the result was written to `submission.csv` (21,600 rows) and uploaded to the Kaggle leaderboard.

	Validation RMSE (May)	Test RMSE (June, Kaggle)
TA baseline (LinearRegression, weather)	29.388	24.246
Our final model	13.425	**11.685**

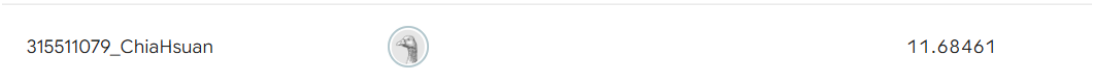


Figure 5: Kaggle leaderboard entry (team `315511079_ChiaHsuan`): test RMSE 11.68461, well below the TA baseline of 24.2461.

Was the validation RMSE a reliable estimate? Yes, and it was slightly conservative. The test RMSE (11.68) is 1.74 lower than the validation RMSE (13.42), and both are far below the corresponding baseline errors. The ranking of models was therefore preserved.

The difference has two causes.

- **June is an easier month for every model.** The TA baseline also improved, from 29.39 on May to 24.25 on June, a drop of 17.5%. Our model dropped by 13.0%, which is consistent with this month-level difference.
- **The final model was retrained with May included.** It was fitted on all of Jan–May, so it had one more month of recent, warm-season data than the model evaluated on May.

The selection on 13 candidates could have made the validation score optimistic. However, the June distribution shift (about 5× more rain in the plum-rain season, warmer weather, and the Dragon Boat Festival on Fri 19 June) did not hurt. The model's error is driven mainly by station-by-hour commuting structure, which is stable from month to month, rather than by weather. A single held-out month can only estimate the error level to within a couple of RMSE points.

Summary

A weather-only linear model explains about 10% of the variance in hourly YouBike demand. Modelling $\log(1+y)$ and adding one-hot time features and crossed station × hour × workday indicators makes a regularised linear model reach a validation RMSE of 13.4 ($R^2 = 0.81$). Scaling leaves OLS unchanged, alters ridge through the effective penalty, and determines whether gradient descent converges at all. Learning curves identified a high-bias model (fixed with richer features) and a high-variance model (fixed with a tuned ridge penalty). The final model reaches a Kaggle test RMSE of 11.68, compared with the TA baseline of 24.25.